

OPERATING SYSTEM AND SYSTEM PROGRAMMING

CODE :25CS2104E

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WEEK-3 PRACTICAL

3. Develop a C program using `fork()` that creates a parent and child process. Display the Process ID (PID), Parent Process ID (PPID), and process states at different stages of execution.

Design an experiment to observe process state transitions (Ready, Running, Waiting, Terminated) using Linux monitoring tools such as `ps`, `top`, and `/proc`. Document the observations.

```
#include <stdio.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/wait.h>

int main() {
    pid_t pid;

    printf("Parent process started\n");
    printf("Parent PID : %d\n", getpid());
    printf("Parent PPID : %d\n", getppid());

    pid = fork();

    if (pid < 0) {
        printf("Fork failed\n");
        return 1;
    }

    if (pid == 0) {
        // Child process
        printf("\nChild process\n");
        printf("Child PID : %d\n", getpid());
        printf("Child PPID : %d\n", getppid());

        printf("Child is running...\n");
        sleep(10);

        printf("Child process terminating...\n");
    }
    else {
        // Parent process
        printf("\nParent process\n");
        printf("Parent PID : %d\n", getpid());
        printf("Child PID : %d\n", pid);

        printf("Parent is waiting for child...\n");
        wait(NULL);

        printf("Child terminated.\n");
        printf("Parent process terminating...\n");
    }

    return 0;
}
```

```
vsainikhil@NIKHIL-BOOK:~$ vi program3.c
vsainikhil@NIKHIL-BOOK:~$ gcc program3.c -o program3
vsainikhil@NIKHIL-BOOK:~$ ./program3
-bash: ./program3: No such file or directory
vsainikhil@NIKHIL-BOOK:~$ ./program3
Parent process started
Parent PID : 618
Parent PPID : 398

Parent process
Parent PID : 618
Child PID : 619
Parent is waiting for child...

Child process
Child PID : 619
Child PPID : 618
Child is running...
Child process terminating...
Child terminated.
Parent process terminating...
vsainikhil@NIKHIL-BOOK:~$ |
```